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Geology 1994;22;933-936
doi: 10.1130/0091-7613(1994)022<0933:SARPFT>2.3.CO;2

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South American reference pole for the mid-Cretaceous: Further constraints in the interpretation of Andean paleomagnetic data

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ABSTRACT

A paleomagnetic study of the Aptian Cerro Barcino Formation in central Patagonia showed the presence of a stable magnetization with a positive tilt test. The obtained pole position (lat 85°S, long 181°E, α_{95} = 5.6°, N = 11) differs from reliable Early and Late Cretaceous South American cratonic poles. The Cerro Barcino pole overlaps coeval poles from the Chilean Andean fore arc previously interpreted to record crustal block rotations. This new result indicates that previously published Andean tectonic models must be reviewed.

INTRODUCTION

Paleomagnetism is a powerful tool in identifying crustal rotations about any axis and understanding response to tectonic stress. Paleomagnetic measurements have been used in the past decade in the study of deformation along the active margin of South America. One difficulty encountered in interpreting these studies is the poorly defined apparent polar wander path for cratonic South America, in particular for the Cretaceous (e.g., Hartley et al., 1988; Beck, 1988). Only two Cretaceous poles are indisputably considered to be reliable reference poles for the continent: the Early Cretaceous Serra Geral pole (Ernesto and Pacca, 1988) and the Late Cretaceous Patagonian basalt pole (Butler et al., 1991). The lack of reliable mid-Cretaceous reference poles has critical implications in the interpretation of Andean paleomagnetic data because (1) there is a widespread presence of mid-Cretaceous rocks in the Chilean Andes and (2) the magnitude of apparent block rotations previously defined in many areas appears to be very small.

The results of a paleomagnetic study in the mid-Cretaceous Cerro Barcino Formation bear on the tectonic implications of Andean paleomagnetic data.

GEOLOGIC SETTING AND SAMPLING

The Cerro Barcino Formation (Codignotto et al., 1978) comprises more than 200 m of lacustrine and fluvial deposits that crop out in the southern end of the Somuncura massif, central Patagonia (Fig. 1). These rocks unconformably overlie volcanic and sedimentary rocks of Middle Jurassic–Early Cretaceous age, and underlie Tertiary deposits.

We collected two oriented block samples or three to four oriented drill cores from selected siltstone, claystone, and sandstone beds for paleomagnetic studies. We defined 11 sites according to changes in sedimentological characteristics along the whole section (fluvial, lacustrine, degree of bioturbation, development of paleosols). Five sites (LP sites, Fig. 1) are located in the eastern foothills of the Sierra Lonco Trapial, where the entire sequence was variably tilted by early Miocene movement. Six sites (CH sites, Fig. 1) are located in the eastern area, where the Cretaceous strata are subhorizontal and have excellent exposures along the Chubut River canyon.

Green tuffs exposed in the Chubut River canyon include a well-preserved assemblage of fossil charophytes and ostracods which constitutes the *Flabellochara cf. harrissi* zone of Aptian age (Musacchio, 1993).

PALEOMAGNETISM

We treated 207 specimens mainly by stepwise thermal demagnetization. Alternating-field demagnetization and chemical leaching also were used in a few cases. Unblocking temperatures between 560 and 590 °C and saturation values of isothermal remanent magnetization between 250 and 500 mT suggest (Ti-poor) magnetite as the

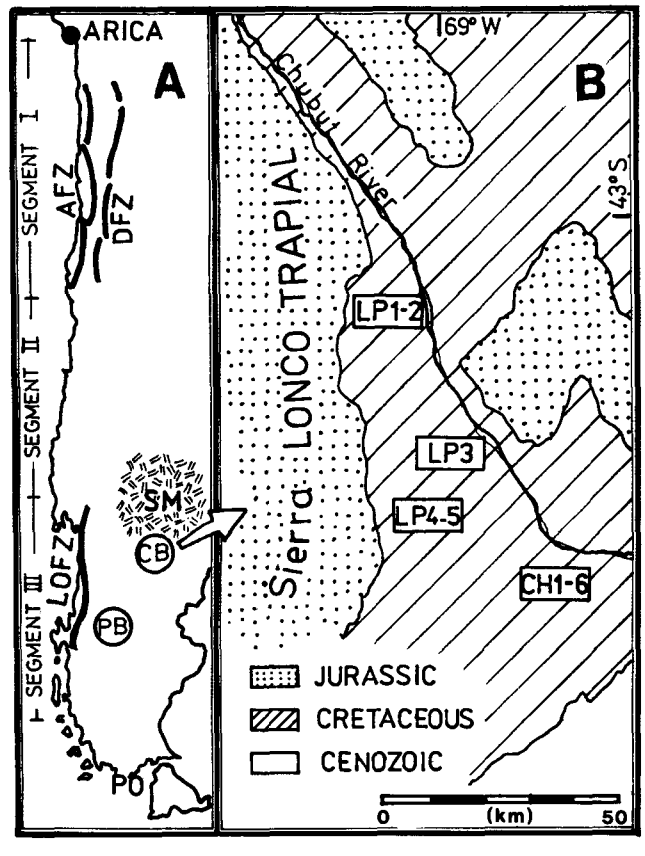


Figure 1. A: Location of main tectonic features and localities mentioned in text. SM is Somuncura massif; PO is Patagonian orocline; PB and CB are Patagonian basalts and Cerro Barcino sampling localities; AFZ, DFZ, and LOFZ are Atacama, Domeyko, and Liquiñe-Ofqui fault zones. Chilean Andean segments (see text) are also shown. B: Sketch map showing Cerro Barcino Formation sampling area. Locations of sites (as listed in Table 1) are also shown.

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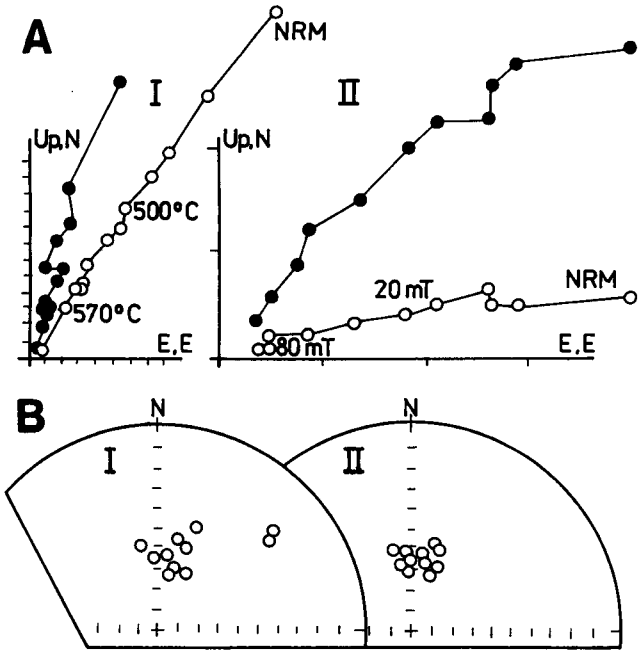


Figure 2. A: Typical demagnetization diagrams of rocks from Cerro Barcino Formation. I: Thermal demagnetization of one specimen from site CH3. II: Alternating field demagnetization of one specimen from site LP1. Both plots have in situ coordinates. Open symbols correspond to vertical plane, solid symbols to horizontal plane. Axes are marked for 1 mA/m intervals. B: Site mean direction before (I) and after (II) tilt correction.

main carrier of the remanence in most of the rock types. In some cases unblocking temperatures up to 640 °C and the absence of saturation until 3 T showed the presence of hematite.

Characteristic directions were determined by principal-components analysis (Kirschvink, 1980) after selection of linear segments by visual inspection of Zijderveld plots. In general the rocks showed a linear demagnetization path trending toward the origin, permitting the isolation of a single (normal polarity) component of magnetization (Fig. 2A), which was consistent in all rocks types. Specimens that showed (1) magnetic instability before removal of 80% of the natural remanent magnetization, (2) changes in susceptibility during thermal treatment, or (3) a poorly defined demagnetization trajectory (MAD value greater than 10°), were rejected from further analysis.

An examination of the directional data shows an improvement of clustering after bedding tilt correction (Fig. 2B). The corresponding increase in the precision parameter (k) implies a positive tilt test, significant at the 99% confidence level based on the *f*-ratio criterion (McElhinny, 1964), that indicates acquisition of the characteristic magnetization prior to the early Miocene. This indication, combined with the presence of normal and reversed polarities in Upper Jurassic–Lower Cretaceous sedimentary rocks in the study area (unpublished data) argues against remagnetization after deposition of the Cerro Barcino Formation, and strongly suggests that the isolated characteristic remanence is of primary origin, associated with deposition or early diagenesis.

A mean pole position for the Cerro Barcino Formation (lat 85°S, long 181°E, α_{95} = 5.6°, k = 66) was calculated using 11 site-mean magnetizations converted to virtual geomagnetic poles (Table 1).

Angular standard deviation is low (10°), possibly because low depositional rates may have caused paleosecular variations to be partially averaged within the site.

The tectonic quiescence in the eastern sampling region (CH sites) together with the arguments mentioned above suggest that the Cerro Barcino pole represents a reliable mid-Cretaceous pole position for cratonic South America. This is also supported by the fact that the Cerro Barcino pole agrees well with coeval African poles (as selected by Hargraves, 1989) after appropriate rotation to African coordinates.

DISCUSSION AND CONCLUSIONS

Several first-order tectonic models have been proposed to account for the Andean rotation pattern defined by paleomagnetism (counterclockwise rotations in Peru and northernmost Chile, and clockwise rotations farther south). Isacks (1988) suggested that forearc oroclinal bending around the Arica zone (Fig. 1A) originated by

TABLE 1. PALEOMAGNETIC RESULTS

Site	N(N$_0$)	In situ Dec.	In situ Inc.	Paleohorizontal* Dec.	Paleohorizontal* Inc.	α_{95}	k	Strike/Dip	VGP†
LP1	6 (6)	51.5	-25.3	359.2	-60.0	4.6	208	355/58	98/87
LP2	6(10)	52.9	-29.1	352.6	-62.3	6.4	144	355/58	19/85
LP3	5(14)	22.2	-46.4	355.4	-64.3	4.6	280	325/25	342/86
LP4	8(12)	14.5	-68.3	12.2	-62.4	3.9	205	093/06	206/81
LP5	5 (8)	27.5	-67.2	22.8	-61.6	11.5	45	093/06	206/73
CH1	5 (8)	349.3	-56.1	349.3	-56.1	7.5	105	Shz	59/79
CH2	5 (9)	18.6	-66.4	18.6	-66.4	6.1	158	Shz	229/76
CH3	4 (8)	14.9	-53.0	14.9	-53.0	4.4	441	Shz	165/74
CH4	6 (9)	7.9	-59.0	7.9	-59.0	6.2	119	Shz	169/83
CH5	9(12)	358.9	-61.3	358.9	-61.3	4.1	156	Shz	82/88
CH6	7(10)	18.5	-55.3	18.5	-55.3	7.5	66	Shz	178/74
Full vector formation means:									
In situ	11	23.3	-55.3			11.0	18		
Paleohor.*	11			6.5	-60.6	4.0	131		
Pole#: 84.9°S, 180.8°E, α_{95} =5.6°, K=66, N=11									

Note: N(N$_0$) are number of samples used in statistics (number of collected samples); Dec. and Inc. are site-mean declination and inclination; α_{95} is confidence level of site mean direction; k is dispersion parameter.
*Paleohorizontal = bedding planes rotated to horizontal about strike (Shz are subhorizontal strata).
†VGP = sites' virtual geomagnetic poles (east longitude/south latitude).
#Formation's pole position.

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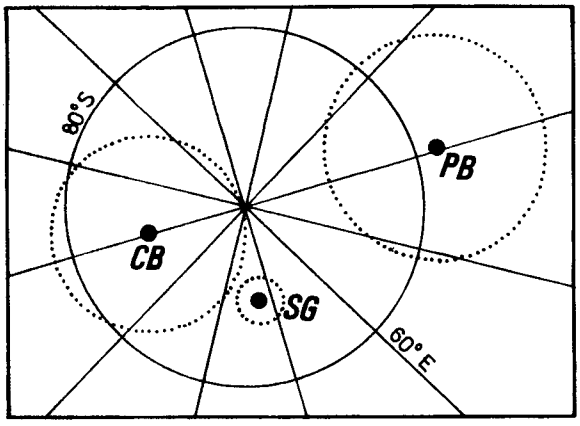


Figure 3. Serra Geral (SG), Cerro Barcino (CB), and Patagonian basalt (PB) poles; 95% confidence intervals are also shown.

differential along-strike shortening in the Central Andes during the late Cenozoic. In contrast, Beck (1988) proposed in situ block rotations in response to the margin-parallel component of an oblique subduction with opposite obliquity north and south of Arica. An alternative explanation, involving both mechanisms, was proposed by Beck et al. (1994). In all these hypotheses the significance of the Arica deflection is fundamental.

The Cerro Barcino pole can be distinguished at 95% confidence level from Lower and Upper Cretaceous reference poles for South America (Fig. 3), recording a previously undocumented mid-Cretaceous pole position. This new reference pole allows a reevaluation of the tectonic significance of the Cretaceous Chilean poles, analyzed in the context of three different tectonic segments of the Chilean Andes (Figs. 1A and 4). Each segment was defined according to its principal paleomagnetic signature: segment I (from Arica elbow to 28°–30°S), segment II (from 28°–30°S to 40°S), and segment III (from 40°S to the Patagonian orocline).

Anomalous declinations are clearly defined in segment I (Fig. 4). These and similar results from pre-Cretaceous rocks (Beck et al., 1994) show widespread occurrence of rotations in this segment.

In segment II, most Cretaceous paleomagnetic poles are roughly concordant. The only significant rotation occurred in the San Fernando area (Beck et al., 1986a), which could be linked to a local deflection in the Andean trend (Irwin et al., 1987). The lack of widespread rotation in this segment is also supported by concordant Jurassic magnetizations (Irwin et al., 1987).

In segment III, rotations were linked to dextral motion in the Liquiñe-Ofqui fault zone (Fig. 1A). Near the fault these rotations show a complex counterclockwise and clockwise pattern (García et al., 1988), which suggests a strong control by fault geometry (Cembrano and Hervé, 1993). Away from the fault zone the only available Cretaceous pole is from rocks in the Alto Palena area (Cembrano et al., 1992), and it is coeval and concordant with the Cerro Barcino pole. This suggests that most deformation associated with the Liquiñe-Ofqui fault zone is near the fault trace, where a belt of cataclastic and mylonitic rocks is present. It is interesting to note that in this segment the only available Paleozoic pole appears to be concordant with the cratonic ones (Beck et al., 1991).

Available Andean paleomagnetic data from Chile show a change from rotated to mostly nonrotated areas around 28°–30°S, whereas in the southern Andes rotations appear to be close to the Liquiñe-Ofqui fault zone and related to its geometry.

A single fore-arc rigid-body rotation (Isacks, 1988) can't explain the variable magnitude (Fig. 4) and timing (Hartley et al., 1992; Riley et al., 1993) of rotations in segment I, which also has nonrotated areas (Jesinkey et al., 1987). Small-block rotations in response to distributed shearing along this segment appear to be a more appropiate explanation.

Oblique subduction with development of trench-linked right-lateral faults (Beck, 1988) can be a way to produce dextral shear along the Chilean margin. However, if a single oceanic plate interacted with western South America during Late Cretaceous and Tertiary time (Pardo Casas and Molnar, 1987), such an oblique subduction model would require additional constraints to explain different behavior along strike of the Chilean Andes (Fig. 4).

Segmentation of the Chilean Andes can also be distinguished by its tectonic evolution (Mpodozis and Ramos, 1990); old sialic basement occurs beneath northern Chile, whereas mainly oceanic crust underlies central Chile (Ramos, 1988). These differences in the nature of the leading edge of South America may determine differential along-strike resistence to shear in the overriding plate (Jarrard, 1986). However, whereas Scheuber and Reutter (1992) proposed right-lateral motion for the Atacama and Domeyko fault zones (Fig. 1A), most geologic evidence indicates left-lateral motion in these fault zones during Cretaceous and Tertiary time (e.g.,

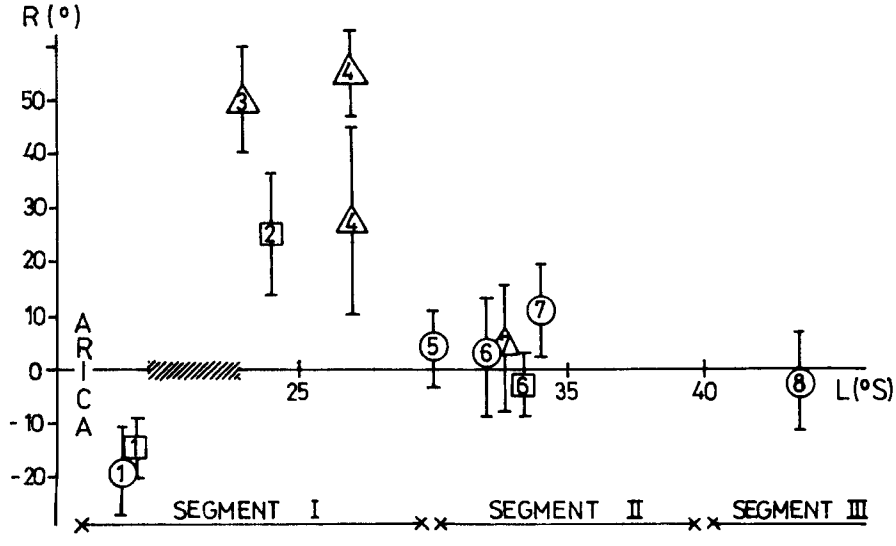


Figure 4. Paleomagnetic rotations of Cretaceous rocks from Chilean Andes. R is rotation (clockwise positive) calculated by method of Beck et al. (1986b). L is latitude of sampling sites. Symbols indicate reference pole used in each case: Serra Geral pole (squares), Cerro Barcino pole (circles), and Patagonian basalt pole (triangles). Numbers in symbols indicate source of Chilean paleomagnetic data: (1) Heki et al. (1983); (2) Hartley et al. (1988); (3) Hartley et al. (1992*); (4) Riley et al. (1993*); (5) Palmer et al. (1980); (6) Beck et al. (1990); (7) Beck et al. (1986a); (8) Cembrano et al. (1992). Bars show 95% confidence intervals. Hatch marks depict Andean zone without any paleomagnetic data, as mentioned in text. Asterisks indicate that magnetization reported in these works could be early Tertiary, despite rocks being of Late Cretaceous age.

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Armijo and Thiele, 1990; Tomlinson et al., 1993), opposite to that required by models involving oblique subduction (Beck, 1988; Beck et al., 1994). This last observation suggests that a shear-driven mechanism should be reconsidered as an explanation for the observed rotations.

Available paleomagnetic and geologic information imposes important constraints on tectonic models previously proposed to account for rotations along the Andes, as noted by Hartley et al. (1992).

A fundamental key for any Andean tectonic model is the location of the counterclockwise-clockwise crustal rotation boundary. This may occur in segment I, in a zone where there is a 400 km gap in paleomagnetic data (Fig. 4). The possible existence of a crustal discontinuity in that zone (e.g., Wörner et al., 1992) suggests that basement characteristics could be related to the Andean rotation pattern.

The nature and preexisting structure of the leading edge of South America could have played an important role in controlling different responses to stress. A more complete (in terms of structure and time) paleomagnetic data set of the Andes, together with a better definition of the South American apparent polar wander path, can help to clarify the origin of Andean crustal rotations.

ACKNOWLEDGMENTS

Supported by the Universidad de Buenos Aires and the Consejo Nacional de Investigaciones Cientificas y Tecnicas (Argentina). I thank my father, H. Somoza, for help in the field; E. Viviani for preparation of illustrations; A. Rapalini, H. Vizán, V. Ramos, C. Siano, and S. Kay for helpful comments; and M. Beck, A. Grunow, and J. Geissman for constructive reviews. Most results were analyzed by the program MAG88 (E. Oviedo).

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Manuscript received April 4, 1994
Revised manuscript received May 25, 1994
Manuscript accepted July 19, 1994